

Akmal Hameed

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SUMMARY

Full-stack engineer with a backend focus, building production AI systems at a recruitment SaaS startup since 2022. Over three years, owned an AI interview and assessment engine on LangGraph, a multi-tenant RAG candidate-matching pipeline on MongoDB Atlas Vector Search with Voyage AI reranking, a tiered Stripe billing platform, and a self-hosted agent observability dashboard. Comfortable driving features from infrastructure to product: Python and FastAPI on the backend, React on the front, Kubernetes and AWS in between. Known for shipping complex agentic and async systems independently, taking ownership of technical decisions, and raising the quality bar across the team.

SKILLS

- **Languages & Backend:** Python, FastAPI, MongoDB, PostgreSQL, GraphQL, REST APIs, Bash, NoSQL
- **Frontend:** React (TypeScript), Next.js, Vue.js, Tailwind CSS, Bootstrap, JavaScript
- **AI / ML:** LangGraph, LLMs, RAG, AI Agents, MongoDB Atlas Vector Search, Voyage, Prompt Engineering
- **DevOps & Cloud:** Kubernetes, Docker, Helm, Kops, AWS, Grafana, Prometheus, GitHub CI/CD
- **AWS:** EC2, S3, SQS, SES, Lambda, OpenSearch, ECS, IAM, Amplify, SageMaker

EXPERIENCE

- **Software Engineer (HiAcuity)** *Sept 2024 – Present*
 - **AI Interview & Assessment Agent (LangGraph):** LangGraph interview engine with competency-block questioning and a fairness-critic node that screens follow-ups for bias before they reach the candidate. Scores write to a structured assessment store.
 - **Multi-Tenant RAG Candidate Search (MongoDB Atlas + Voyage):** Multi-tenant RAG pipeline on MongoDB Atlas Vector Search with chunked, max-pooled embeddings and Voyage AI reranking; a reasoning step routes structured filters (experience, location, dates) separately from semantic search. Meaningfully improved match precision over keyword-only search across **100K+ candidate profiles** with sub-second latency.
 - **Stripe Subscription and Billing System:** Owned the full build of a feature-gated subscription system on Stripe, launching **3 pricing tiers** and eliminating nearly all manual billing exceptions through webhook-driven access control automation.
 - **Live Interview Transcription & Validation (Recall + WebSockets):** Built a WebSocket pipeline on Recall streaming live transcription from candidate interviews. Validates spoken claims against the candidate's resume and general fact-checks in real time, surfacing deeper follow-up questions and validation flags to recruiters as the interview happens.
 - **Frontend Engineering:** Reworked React component architecture for the candidate portal, introducing useReducer and useContext patterns alongside custom hooks that cut unnecessary re-renders and simplified state management across the app.
 - **Engineering Quality and Mentorship:** Set the code review standard for the team, running structured reviews and technical walkthroughs that measurably improved PR quality and reduced back-and-forth across the board.
- **Junior Software Engineer (HiAcuity)** *Mar 2023 – Sept 2024*
 - **Microservice Standardization:** Identified that shared logic had drifted across 6 microservices with no consistency, then ran a cross-service refactor to consolidate utilities and base classes. Cut duplication significantly and made onboarding new engineers faster.
 - **Cloud Cost Reduction:** Dug into production workloads with Prometheus and Grafana, right-sized EC2 instance classes, and tightened auto-scaling policies, lowering monthly infra spend with no impact on uptime or performance.
- **Software Engineering Intern (HiAcuity)** *Sept 2022 – Mar 2023*
 - **Test Automation and QA:** Built **40+ Selenium test cases** covering critical UI flows, cutting manual regression effort considerably. Ran load tests with JMeter and Datadog and flagged **3 performance bottlenecks** before they reached production.
 - **Secure API Development:** Built FastAPI endpoints with role- and group-based access control for two user personas: candidates and recruiters. The implementation held up through a full security review with no access control issues found.

PROJECTS

- **AI Resume Search: Candidate Matching Engine:** Multi-stage retrieval pipeline where LangGraph routes queries through Pinecone vector search, then applies structured MongoDB filters on experience, skills, and location before surfacing results. Tested against **10K+ resumes** with consistently strong top-3 matches. **Stack:** LangGraph, Pinecone, MongoDB, Python, LLMs
- **HouseDiffusion: Generative Floor Plan Generator:** Fine-tuned a Diffusion and Transformer model for room-level layout synthesis. Generation runs in **under 5 seconds**. Users work through a React drag-and-drop canvas; FastAPI handles model inference on the backend. **Stack:** PyTorch, ReactJS, React Flow, FastAPI
- **Agentic-Map: Multi-Agent Trip Resolver:** Multi-agent LangGraph trip planner that resolves vague natural-language stops (e.g. "grab a coffee on the way"). A supervisor graph runs parallel Scout agents per stop alongside a Navigator (time budget) and Concierge (weather/ETA), with human-in-the-loop approval and checkpointed, resumable sessions. **Stack:** LangGraph, Claude (langchain-anthropic), Google Maps Routes v2 / Places v1, Open-Meteo, Python
- **Assignment Reminder: Automated LMS Tracker:** Selenium-based deadline tracker that fires WhatsApp reminders at **10, 3, and 1 day** out. Deployed on EC2 with GitHub Actions CI/CD. Picked up by **30+ students** in its first semester without any promotion. **Stack:** Python, Selenium, AWS EC2, GitHub Actions
- **DevTools MCP Server: GitHub and CI Integration for Claude:** MCP server that exposes GitHub PRs, CI run logs, and issue trackers to Claude via typed tool schemas. Lets engineers run natural-language code reviews and draft release notes without context-switching out of the conversation. **Stack:** Python MCP SDK, Claude API, GitHub REST API, FastAPI, Docker
- **AgentTrace: LLM Agent Observability Dashboard:** Built this because LangSmith does not cover self-hosted pipelines well. Hooks into LangGraph callbacks to capture node execution times, tool call sequences, and per-step token usage; stores trace history in PostgreSQL for cross-run comparisons. React frontend renders live traces as collapsible execution trees via WebSocket push; Prometheus tracks latency and cost. **Stack:** LangGraph, FastAPI, WebSockets, React, Redis, PostgreSQL, Prometheus

EDUCATION

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| • APIIT Sri Lanka (Staffordshire University) | Kandy, Sri Lanka |
| • <i>B.Sc. in Computer Science (Top-Up), First Class Honours</i> | <i>Mar 2023 – Mar 2024</i> |
| • BCAS Sri Lanka (Pearson) | Kandy, Sri Lanka |
| • <i>HND in Computer Science, Merit</i> | <i>Dec 2021 – Sept 2022</i> |